

MPC-MAP Assignment No. 3 - Report

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Task 1

I implemented `predict_pose` using a standard differential drive kinematic model. To increase particle variance, I added Gaussian noise (`randn() * 1.3`) to the wheel velocities, allowing the particles to properly explore the state space.

Task 2

`compute_lidar_measurement` uses the `ray_cast` function to calculate distances to the closest walls for all 8 Lidar channels. In `weight_particles`, I calculated the Mean Squared Error (MSE) between simulated and real measurements (ignoring NaN values) and assigned weights using a Gaussian function.

Task 3

I used the "Roulette Wheel" selection method in `resample_particles`. By generating a random number against a cumulative sum of normalized weights, highly-weighted particles are duplicated while poor ones are discarded.

Task 4

I randomly initialized 150 particles across the map. To ensure reliable convergence, I added a spin to localize behavior in `plan_motion`. The robot spins in place for the first 50 iterations, allowing the particles to form a tight cluster around the true pose before path-following begins.

The most critical parameters are the motion noise and Lidar tolerance (sigma). I used a high motion noise (1.3) so particles could keep up with the robot's movement. The sigma value dictates the precision-stability trade-off: as shown in Figure 1, a value of $\sigma = 0.75$ provides robust tracking where the particle cluster is stable, though the estimated path is slightly rougher. Conversely, Figure 2 demonstrates that $\sigma = 0.25$ is too strict, causing the filter to lack tolerance for sensor noise and completely lose the robot.

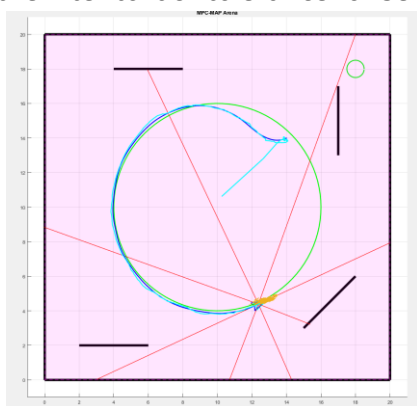


Figure 1 – Sigma 0.75

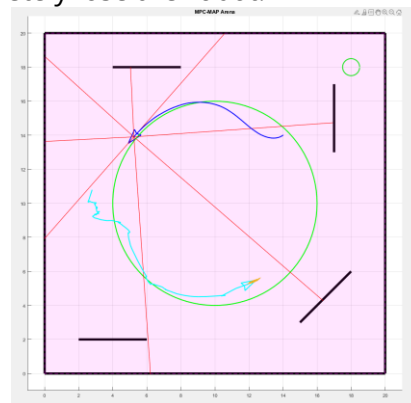


Figure 2 – Sigma 0.25